

Mathematical Methods I

Directional Derivatives and Vector Fields

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1. Derivative with Respect to a Tangent Vector

A **directional derivative** measures the instantaneous rate of change of a scalar function as one moves in a given direction. It generalizes partial derivatives: these describe change along the coordinate axes, whereas a directional derivative allows us to consider any vector.

Definition 1.1 (Directional Derivative)

Let $\Omega \subseteq \mathbb{R}^n$ be an open set, $p \in \Omega$, and $\mathbf{v} \in \mathbb{R}^n$. The **tangent vector** $\mathbf{v}_p$ is the vector $\mathbf{v}$ based at the point p . For $f : \Omega \rightarrow \mathbb{R}$, define

$$\mathbf{v}_p[f] = \left. \frac{d}{dt} f(p + t\mathbf{v}) \right|_{t=0} = \lim_{t \rightarrow 0} \frac{f(p + t\mathbf{v}) - f(p)}{t}, \quad (1)$$

provided the limit exists. This is also written as $D_{\mathbf{v}}f(p)$.

The curve $\gamma(t) = p + t\mathbf{v}$ traces a straight line when $\mathbf{v} \neq \mathbf{0}$, with $\gamma(0) = p$ and velocity $\gamma'(0) = \mathbf{v}$. Thus, $\mathbf{v}_p[f]$ is the ordinary derivative of the single-variable function $f \circ \gamma$ at $t = 0$.

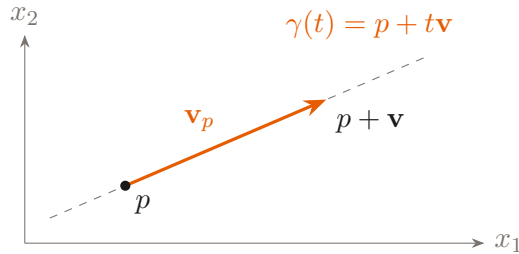


Figure 1: The derivative is evaluated at p along the line with velocity $\mathbf{v}$.

2. Computation in Cartesian Coordinates

Theorem 2.1 (Coordinate Formula)

If $f : \Omega \rightarrow \mathbb{R}$ is differentiable at p and $\mathbf{v} = (v_1, \dots, v_n)$, the chain rule gives

$$\mathbf{v}_p[f] = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i}(p) = \nabla f(p) \cdot \mathbf{v}. \quad (2)$$

In particular, in $\mathbb{R}^3$,

$$\mathbf{v}_p[f] = v_1 \frac{\partial f}{\partial x_1}(p) + v_2 \frac{\partial f}{\partial x_2}(p) + v_3 \frac{\partial f}{\partial x_3}(p). \quad (3)$$

Here $\nabla f(p)$ is the **gradient**: the vector of partial derivatives of f evaluated at p .

The differentiability assumption is important: the mere existence of partial derivatives at p is not enough to justify this formula.

Direction and Speed

$\mathbf{v}_p[f]$ is defined for a vector of any length; it measures change per unit of the parameter t . To measure change per unit distance in the direction of $\mathbf{v} \neq \mathbf{0}$, use the unit vector $\hat{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|$:

$$D_{\hat{\mathbf{v}}}f(p) = \frac{\mathbf{v}_p[f]}{\|\mathbf{v}\|}. \quad (4)$$

Example 2.2 (Evaluation at a Point)

Let $f(x, y, z) = x^2y + z^2$, $p = (1, 2, -1)$, and $\mathbf{v} = (2, -1, 3)$. Then

$$\nabla f(x, y, z) = (2xy, x^2, 2z), \quad \nabla f(p) = (4, 1, -2).$$

Therefore,

$$\mathbf{v}_p[f] = 4(2) + 1(-1) + (-2)(3) = 1.$$

This can be verified directly along the line:

$$f(p + t\mathbf{v}) = (1 + 2t)^2(2 - t) + (-1 + 3t)^2 = 3 + t + 13t^2 - 4t^3.$$

Its derivative at $t = 0$ is 1. Since $\|\mathbf{v}\| = \sqrt{14}$, the derivative in the corresponding unit direction is $1/\sqrt{14}$.

3. Properties of the Directional Derivative

Let $f, g : \Omega \rightarrow \mathbb{R}$ be differentiable at p , let $\mathbf{v}_p, \mathbf{w}_p$ be tangent vectors at the same point, and let $a, b \in \mathbb{R}$ be constants. The directional derivative satisfies the usual rules of differentiation.

Theorem 3.1 (Linearity)

Linearity in the function:

$$\mathbf{v}_p[af + bg] = a \mathbf{v}_p[f] + b \mathbf{v}_p[g]. \quad (5)$$

Linearity in the vector:

$$(a\mathbf{v}_p + b\mathbf{w}_p)[f] = a \mathbf{v}_p[f] + b \mathbf{w}_p[f]. \quad (6)$$

The second identity follows from the linearity of the differential $df_p(\mathbf{v}) = \nabla f(p) \cdot \mathbf{v}$. It applies to vectors without normalizing them after addition.

Theorem 3.2 (Product Rule or Leibniz Rule)

$$\mathbf{v}_p[fg] = \mathbf{v}_p[f] g(p) + f(p) \mathbf{v}_p[g]. \quad (7)$$

The undifferentiated factors are evaluated at the base point p .

Theorem 3.3 (Chain Rule)

If $h : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at $f(p)$, then

$$\mathbf{v}_p[h \circ f] = h'(f(p)) \mathbf{v}_p[f]. \quad (8)$$

This follows by differentiating $h(f(p + t\mathbf{v}))$ with respect to t at $t = 0$.

Example 3.4 (Applying the Rules)

For f, p , and $\mathbf{v}$ from Example 2.2, $f(p) = 3$ and $\mathbf{v}_p[f] = 1$. By the chain rule,

$$\mathbf{v}_p[f^2] = 2f(p) \mathbf{v}_p[f] = 6.$$

If, in addition, $g(x, y, z) = x + z$, then $g(p) = 0$ and $\mathbf{v}_p[g] = 2 + 3 = 5$. The Leibniz rule yields

$$\mathbf{v}_p[fg] = 1(0) + 3(5) = 15.$$

4. Vector Fields as Operators

A **vector field** V assigns a tangent vector $V(p)$ to each point $p \in \Omega$. Unlike a fixed vector $\mathbf{v}_p$, which acts at a single point, the field allows us to differentiate a function at every point of its domain.

Definition 4.1 (Action of a Field on a Function)

For a differentiable function $f : \Omega \rightarrow \mathbb{R}$, the expression $V[f]$ denotes the new **scalar function** defined by

$$(V[f])(p) = V(p)[f]. \quad (9)$$

If $V(p) = (v_1(p), \dots, v_n(p))_p$, then

$$(V[f])(p) = \sum_{i=1}^n v_i(p) \frac{\partial f}{\partial x_i}(p). \quad (10)$$

As an identity between functions, this is written as

$$V[f] = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i}. \quad (11)$$

Theorem 4.2 (Properties of the Operator V)

For differentiable functions f, g , constants $a, b \in \mathbb{R}$, and a scalar function ϕ defined on Ω ,

$$V[af + bg] = aV[f] + bV[g], \quad (12)$$

$$V[fg] = V[f]g + fV[g], \quad (13)$$

$$(\phi V)[g] = \phi V[g]. \quad (14)$$

In the last identity, ϕ multiplies the field: it is not differentiated. In contrast, $V[\phi g] = V[\phi]g + \phi V[g]$ when ϕ is differentiable.

Example 4.3 (From a Number to a Function)

In $\mathbb{R}^2$, let $V(x, y) = (y, -x)_{(x, y)}$ and $f(x, y) = x^2 y$. Then

$$V[f](x, y) = y(2xy) - x(x^2) = 2xy^2 - x^3.$$

The result is a function of (x, y) . A number is obtained only after evaluating it at a point: for $p = (1, 2)$,

$$(V[f])(p) = 7 = V(p)[f], \quad V(p) = (2, -1)_p.$$

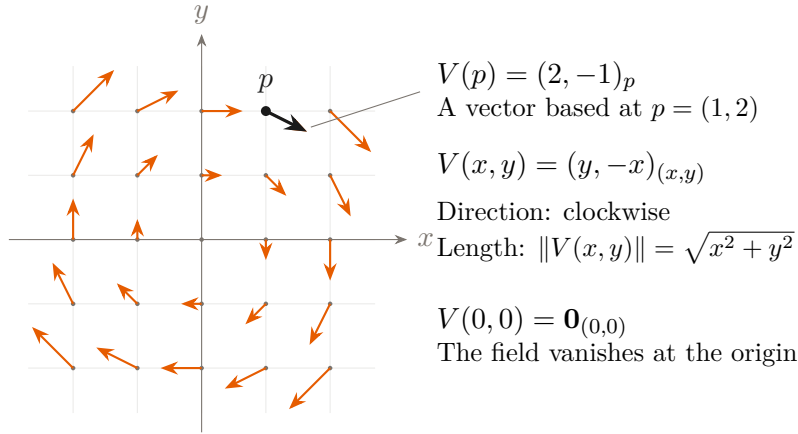


Figure 2: A vector field assigns a vector to every point. For the field in Example 4.3, the vectors are tangent to circles centered at the origin and grow in length with the distance from it. All arrows use the same reduced scale; the highlighted arrow represents the vector at p .

5. The Natural Frame and Partial Derivatives

In Cartesian coordinates, the fields of the **natural frame** $U_1, \dots, U_n$ assign the standard basis vectors to each point: $U_i(p) = (\mathbf{e}_i)_p$. In $\mathbb{R}^3$,

$$U_1(p) = (1, 0, 0)_p, \quad U_2(p) = (0, 1, 0)_p, \quad U_3(p) = (0, 0, 1)_p.$$

Proposition 5.1 (Natural Fields and Partial Derivative Operators)

For every differentiable function f ,

$$U_i[f] = \frac{\partial f}{\partial x_i}. \quad (15)$$

In particular, using coordinates (x, y, z) ,

$$U_1[f] = \frac{\partial f}{\partial x}, \quad U_2[f] = \frac{\partial f}{\partial y}, \quad U_3[f] = \frac{\partial f}{\partial z}.$$

Through its action on functions, U_i is identified with the operator $\partial/\partial x_i$.

Every vector field can be expressed uniquely in terms of its coordinate functions:

$$V = \sum_{i=1}^n v_i U_i = \sum_{i=1}^n v_i \frac{\partial}{\partial x_i}. \quad (16)$$

Applying this expression to f gives:

$$V[f] = \left(\sum_{i=1}^n v_i U_i \right) [f] = \sum_{i=1}^n v_i U_i[f] = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i}. \quad (17)$$

The coefficients are functions, not necessarily constants. For the coordinate functions, $U_i[x_j] = \delta_{ij}$ and hence $V[x_j] = v_j$; this recovers each component and explains uniqueness.

Example 5.2 (One Field, Two Notations)

The field from Example 4.3 can be written as

$$V = yU_1 - xU_2 = y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y}.$$

Thus, $V[x] = y$, $V[y] = -x$, and $V[x^2y] = 2xy^2 - x^3$.

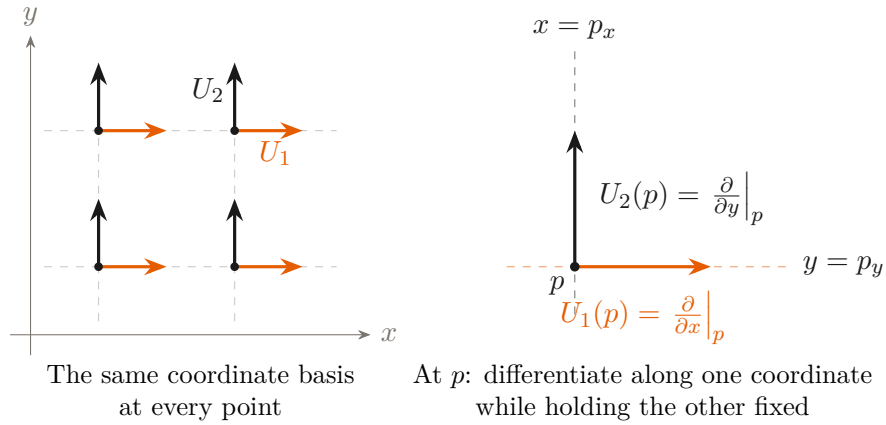


Figure 3: The Cartesian natural frame in $\mathbb{R}^2$. The fields $U_1 = \partial/\partial x$ (orange) and $U_2 = \partial/\partial y$ (black) are tangent to the coordinate lines and form a basis at each point. Their action on a scalar function gives $U_1[f] = \partial f/\partial x$ and $U_2[f] = \partial f/\partial y$. The right panel enlarges one such basis.